



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHY F422
Course Title : Group Theory and Applications
Instructor-in-Charge : Bhavesh Chauhan

1. Course Description:

This is an introductory course on Group Theory, designed for undergraduates and PhD scholars, with an emphasis on applications in physics. Many physical systems exhibit symmetry, which allows for a simplified analysis. The mathematical language of describing and studying a vast class of these symmetries is Group Theory. In each case, the symmetry operations are represented by individual group elements and their combinations. Discrete symmetries entail discrete groups, which may contain a finite or infinite number of elements. Continuous symmetries entail continuous groups (also called as 'Lie groups'). In this course, we will cover the structure, classification, and representations of groups. We will see how group theory can help us understand some aspects of quantum mechanics, classical mechanics, and most importantly, particle interactions.

2. Scope and Objective of the Course:

- Identify and classify discrete and continuous symmetries in physical systems using the foundational concepts of group theory.
- Apply the principles of representation theory to decompose group representations into irreducible components and interpret their physical significance.
- Demonstrate the application of group theory in quantum mechanics by analyzing the symmetry structure of the hydrogen atom and explaining the role of conserved quantities and degeneracies.
- Analyze physical systems under spacetime symmetry transformations using continuous groups such as the Lorentz and Poincaré groups.
- Analyze particle interactions and fundamental forces using the mathematical framework of Lie groups and Lie algebras.

3. Text Books:

T1: *Group Theory in Physics* by Wu-Ki Tung (World Scientific Pub Co Inc)



4. Reference Books:

R1: *Elements of Group Theory for Physicists*, A W Joshi (New Age International Publishers)

R2: *Lie Algebras in Particle Physics*, H. Georgi, (Westview Press)

R3: *Group Theory a Physicist's Survey*, Pierre Ramond, (Cambridge University)

R4: *Group Theory in a Nutshell for Physicists*, A. Zee, (Princeton).

5. Course Plan: In the reference column CN stands for Class Notes.

Module No.	Lecture Session	Ref.	Learning outcomes
1. Overview	L1: Introduction of the course in view of its necessity in the context of fundamental physics; the course structure will be briefed	CN	Explain the relevance of group theory in fundamental physics and outline the structure and goals of the course.
2. Basic concepts	L2—L3: Importance of symmetry in nature and in natural law, groups as the language of symmetry, definition and introduction.	T1, CN	Identify the role of symmetry in physics and explain how groups serve as the underlying mathematical language.
3. Classification and Representation	L4—L7: Subgroups, cosets, invariant and normal subgroups, quotient groups and product groups, Homomorphisms and Isomorphisms. L8—L12: Group representation: unitary representation, reducible and irreducible representation, direct sum and tensor products, character table, Schur's lemmas.	T1, CN	Analyze the structure of groups and understand representation theory.
4. Continuous Groups	L13—L17: Lie groups, infinitesimal transformation, Jacobi identity, structure constants, killing forms, subalgebras; Real and Complex Lie algebra, Irreducible representations of Lie groups and Lie algebras; Adjoint Representations.	T1, R2, CN	Analyze the structure and representations of Lie groups and Lie algebras, with applications to matrix Lie groups.
5. Application to Quantum Systems	L18—L20: Rotations and Angular Momentum, SO(3) and SU(2), Infinitesimal rotations and generators, covering groups, applications, introduction to Spinor representations, Clifford algebra, Dirac matrices, Majorana and Weyl fermions, Representations of SU(3), Pauli vs Gell-Mann matrices	T1, R1, CN	Apply group-theoretic methods to analyze angular momentum, spinor representations, and SU(2)/SU(3) symmetries in quantum systems.
6. Application in Classical Systems	L21—L24: Relativistic symmetries, Lorentz and Poincaré groups, introduction to particle representations, Casimir operators.	T1, R3, R4, CN	Analyze relativistic symmetries using the Lorentz and Poincaré groups and classify particle states via their Casimir operators.
6. Lie groups and particle physics	L25—L38: Introduction to Cartan-Weyl basis, Classification of simple Lie algebras, Roots and weights, Dynkin diagrams, SU(3) representations revisited, Eightfold way, Isospin, Meson and Baryon masses, symmetry breaking, color, introduction to Abelian and Non-Abelian gauge groups	R2, CN	Classify simple Lie algebras using the Cartan-Weyl formalism and apply symmetry principles to particle physics and gauge theories.



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6. Evaluation Scheme:

Component	Duration	Weightage (%) [Marks]	Date & Time	Nature of component (Closed Book/ Open Book)
Term-paper (s)	—	10 [20]	—	—
Quiz (Announced) [Best 3 out of 4]	Approx. 20 min each	20 [40]	TBA	Open Notes
Mid-Semester Test	—	25 [50]	TBA	Closed Book
Comprehensive Examination	180 min	45 [90]	TBA	Closed/Open Book

7. Chamber Consultation Hour: Thursday from 5 pm to 6 pm

8. Notices: All notices will be announced on Google Classroom only

9. Make-up Policy: No make-up for Quiz, Term Paper, and Mid-semester test. Make-up for comprehensive exam will be granted to genuine cases with prior intimation and permission as per institute rules.

10. Note: It is the responsibility of the students to keep all the evaluated and returned answer sheets till the end of the semester.

Bhavesh Chauhan
Instructor-in-charge
PHY F422